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#include <iostream>
#include <cmath>
#include <functional>
using namespace std;
double f(double x) {
    return x * x * x - 10 * x * x - 5 * x + 5;
}
double bisection(double a, double b, double tolerance, int max_iterations) {
    if (f(a) * f(b) >= 0) {
        cout << "The bisection method requires that f(a) and f(b) have different signs." << endl;
        return NAN;
    }
    double c;
    for (int i = 0; i < max_iterations; i++) {
        c = (a + b) / 2;
        if (fabs(f(c)) < tolerance) {
            break;
        }
        if (f(c) * f(a) < 0) {
            b = c;
        } else {
            a = c;
        }
    }
    return c;
}
int main() {
    double a, b, tolerance;
    int max_iterations;

    cout << "Enter the interval [a, b]: ";
    cin >> a >> b;
    cout << "Enter tolerance: ";
    cin >> tolerance;
    cout << "Enter maximum iterations: ";
    cin >> max_iterations;

    double root = bisection(a, b, tolerance, max_iterations);
    if (!isnan(root)) {
        cout << "The root is approximately: " << root << endl;
    }

    return 0;
}

```